

Explicit Two-Sided Bounds for the Backward Shift on H^6

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Source Question

Karlovykh and Shargorodsky ask in Open Problem 5.4 of arXiv:2308.04072 to determine the exact norm of the backward shift

$$(\mathbf{B}f)(z) = \frac{f(z) - f(0)}{z}$$

on H^p for $1 \leq p < \infty$ [1]. The source passage is reproduced in Figure 1.

The problem has subsequently been solved at $p = 1$ by Brevig and Seip [2], and at $p = 4$ by Bampouras and Llinares [3]; of course the values at $p = 2$ and $p = \infty$ were already known. This packet gives a rigorous explicit bracket at the next even exponent $p = 6$. It does not claim the exact H^6 norm.

Definitions

Normalized arclength measure on $\mathbb{T}$ is denoted by m . For $1 \leq p < \infty$,

$$\|f\|_{H^p}^p = \int_{\mathbb{T}} |f(\zeta)|^p dm(\zeta).$$

Write $\varphi = (1 + \sqrt{5})/2$. We use the known sharp values

$$\|\mathbf{B}\|_{H^1} = \frac{2}{\sqrt{3}}, \quad \|\mathbf{B}\|_{H^2} = 1, \quad \|\mathbf{B}\|_{H^4} = \varphi^{1/4}, \quad \|\mathbf{B}\|_{H^\infty} = 2.$$

Proof Intuition

For the lower bound, test the operator on an affine function of a disk automorphism I . Because I is inner, every mixed moment $\int I^j \bar{I}^k$ reduces to a power of $I(0)$. At $p = 6$ this makes both norms exact degree-six polynomials in two rational parameters. A near-optimal rational choice gives the stated exact lower bound.

For the upper bound, the Hardy spaces form the usual complex interpolation scale. Interpolating the new sharp H^4 norm with the sharp H^∞ norm gives a global estimate for $4 \leq p \leq \infty$, and hence the H^6 upper bound. The same argument between the other known endpoints gives the piecewise estimate recorded below.

and provides a simpler proof of (5.4).

5.2. Exact value of the norm of the backward shift operator on H^p

We think that the question about the exact value of $\|I - \mathbf{K}_0\|_{\mathcal{B}(H^p)}$ is particularly interesting, and although it is a special case of Problem 5.2 we state it again below in terms of the backward shift operator

$$(\mathbf{B}f)(e^{i\theta}) := e^{-i\theta} \left(f(e^{i\theta}) - \widehat{f}(0) \right) = e^{-i\theta} ((I - \mathbf{K}_0)f)(e^{i\theta}), \quad f \in H^p.$$

Clearly,

$$\begin{aligned} |\mathbf{B}f| &= |(I - \mathbf{K}_0)f| \implies \|\mathbf{B}f\|_{H^p} = \|(I - \mathbf{K}_0)f\|_{H^p} \text{ for all } f \in H^p \\ &\implies \|\mathbf{B}\|_{\mathcal{B}(H^p)} = \|I - \mathbf{K}_0\|_{\mathcal{B}(H^p)}. \end{aligned}$$

In particular,

$$\|B\|_{\mathcal{B}(H^\infty)} = 2$$

(see (5.3) and [10, Theorem 2.5]).

Open problem 5.4. *Let $1 \leq p < \infty$. Find the exact value of the norm $\|\mathbf{B}\|_{\mathcal{B}(H^p)}$ of the backward shift operator.*

Figure 1: Open Problem 5.4 in the source PDF, page 15.

The Inner-Function Calculation

Lemma 1. *For $0 \leq a < 1$, let*

$$I_a(z) = \frac{a + z}{1 + az}.$$

For real c , define $D(a, c) = \|I_a - c\|_{H^6}^6$. Then

$$\begin{aligned} D(a, c) &= 1 + 9c^2 + 9c^4 + c^6 - 6ac - 18ac^3 - 6ac^5 \\ &\quad + 6a^2c^2 + 6a^2c^4 - 2a^3c^3, \end{aligned}$$

and

$$D(a, a) = (1 - a^2)(a^4 + 4a^2 + 1).$$

Proof. The function I_a is inner and $I_a(0) = a$. Hence for integers $0 \leq j, k \leq 3$,

$$\int_{\mathbb{T}} I_a^j \overline{I_a^k} dm = a^{|j-k|}.$$

Indeed, if $j \geq k$, unimodularity on $\mathbb{T}$ reduces the integrand to I_a^{j-k} , whose mean is $I_a(0)^{j-k}$; the other case follows by conjugation. Expanding

$$D(a, c) = \|(I_a - c)^3\|_{H^2}^2$$

and inserting these moments gives the displayed polynomial. Substitution $c = a$ and factorization gives the second identity. $\square$

Main Partial Result

Theorem 1 (Explicit H^6 bracket). *The backward shift on H^6 satisfies*

$$\left(\frac{6229716996096}{1533633162625}\right)^{1/6} \leq \|\mathbf{B}\|_{H^6} \leq 2^{1/3}\varphi^{1/6}.$$

Numerically,

$$1.2631583709 \leq \|\mathbf{B}\|_{H^6} \leq 1.3651318687.$$

Proof. For the lower bound take

$$a = \frac{32}{35}, \quad c = \frac{11}{28}, \quad f = I_a - c.$$

Since $f(0) = a - c$ and $|z| = 1$ on $\mathbb{T}$,

$$\|\mathbf{B}f\|_{H^6}^6 = \|I_a - a\|_{H^6}^6 = D(a, a).$$

Lemma 1 and exact rational arithmetic give

$$\frac{\|\mathbf{B}f\|_{H^6}^6}{\|f\|_{H^6}^6} = \frac{D(32/35, 32/35)}{D(32/35, 11/28)} = \frac{6229716996096}{1533633162625}.$$

Taking sixth roots proves the lower estimate.

For the upper bound, use the standard complex interpolation identity for Hardy spaces [4, Chapter 7]. Interpolating $\mathbf{B} : H^4 \rightarrow H^4$, whose sharp norm is $\varphi^{1/4}$ by [3], with $\mathbf{B} : H^\infty \rightarrow H^\infty$, whose norm is 2, yields

$$\|\mathbf{B}\|_{H^p} \leq (\varphi^{1/4})^{4/p} 2^{1-4/p} = \varphi^{1/p} 2^{1-4/p}, \quad 4 \leq p \leq \infty.$$

Putting $p = 6$ gives the claimed upper estimate. □

Theorem 2 (Interpolation bounds for all finite exponents). *For $1 \leq p < \infty$,*

$$\|\mathbf{B}\|_{H^p} \leq \begin{cases} (2/\sqrt{3})^{2/p-1}, & 1 \leq p \leq 2, \\ \varphi^{1/2-1/p}, & 2 \leq p \leq 4, \\ \varphi^{1/p} 2^{1-4/p}, & 4 \leq p < \infty. \end{cases}$$

Proof. Interpolate respectively between the sharp endpoint pairs (H^1, H^2) , (H^2, H^4) , and (H^4, H^∞) . Solving $1/p = (1 - \theta)/p_0 + \theta/p_1$ and applying the operator interpolation inequality gives exactly the three displayed exponents. □

Verification and Upgrade Attempts

The script `code/verify_h6_bound.py` independently expands the mixed moments, verifies the factorization of $D(a, a)$, checks the exact rational quotient, and prints the numerical endpoints.

Several routes toward the exact H^6 value were tested before stopping:

1. A current-literature check found the H^1 and May 2026 H^4 solutions, but no H^6 solution.

2. Optimizing the affine-inner family numerically gives $a \approx 0.9146566921$, $c \approx 0.3930991838$, and sixth-power ratio 4.0620742602; the rational pair in Theorem 1 is within 10^{-5} at the sixth-power level.
3. A degree-six harmonic moment-majorant was sought with equality on the candidate circle. The resulting polynomial is divisible by the circle equation only once, not twice, so it cannot be globally nonnegative. Thus the natural scalar/SOS certificate cannot prove exactness.
4. Direct Euler–Lagrange expansion at $p = 6$ introduces three independent mixed correlations; the single sharp Fourier-symmetry lemma that closes the H^4 proof does not control them simultaneously.
5. Optimized finite Taylor polynomials of degrees up to six remained below the affine-inner value and therefore supplied evidence but no upper bound.
6. Powers $(I - c)^k$, $2 \leq k \leq 6$, were numerically optimized and gave strictly smaller ratios.
7. General real polynomials of degree up to four in one inner function did not improve the affine family in bounded numerical searches.
8. Interpolation gives the rigorous upper estimate above, but its equality conditions are incompatible with the known H^4 extremals and do not identify an H^6 extremizer.

These obstructions leave a genuine gap of about 0.102 in operator norm, so the exact H^6 problem remains open in this packet.

Scope and Novelty Check

This is a partial answer to Open Problem 5.4, not a determination of the exact norm. The exact cases currently identified are $p = 1, 2, 4, \infty$. Cheap run indexes were searched for arXiv:2308.04072, backward shift, H^p , and Hardy space norm. Web/arXiv searches on August 11, 2026 used the exact phrases “norm of the backward shift operator on H^p ”, “backward shift on H^4 ”, and “backward shift H^6 exact norm”. They found [2, 3] but no exact H^6 result.

The May 2026 paper [3] announces a forthcoming preprint by Jiménez and Vukotić on general norm estimates. That preprint was not findable by exact title at the time of this check, so overlap with the explicit lower-bound family remains possible. Novelty confidence is therefore moderate; proof validity does not depend on novelty.

Human-Review Recommendation

The core proof is elementary. Review should check the mixed-moment expansion in Lemma 1, the exact rational substitution, and use of the standard Hardy-space interpolation scale including the H^∞ endpoint.

References

- [1] O. Karlovykh and E. Shargorodsky, *Bounded compact and dual compact approximation properties of Hardy spaces: new results and open problems*, arXiv:2308.04072 (2023).

- [2] O. F. Brevig and K. Seip, *The norm of the backward shift on H^1 is $2/\sqrt{3}$* , Pure Appl. Funct. Anal. 9 (2024), 991–994; arXiv:2309.11360.
- [3] K. Bampouras and A. Llinares, *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$* , arXiv:2605.10469 (2026).
- [4] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.